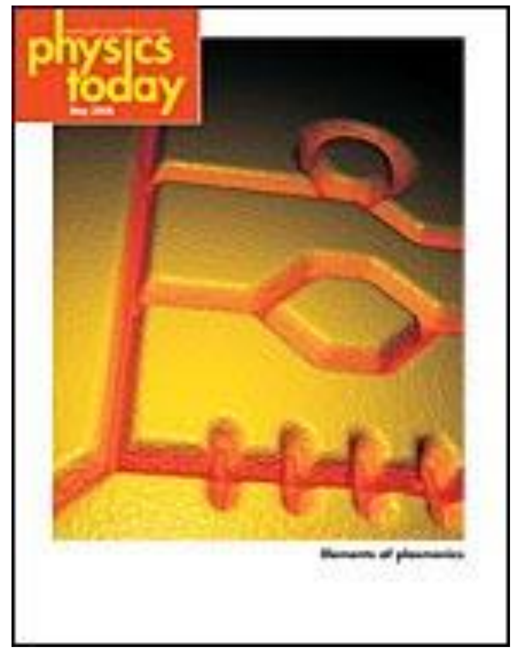
Plasmonic materials and devices

- **Circuit model**
- **Drude model (revision)**
- **Localized Surface Plasmons**
- **Surface Plasmon Polaritons**

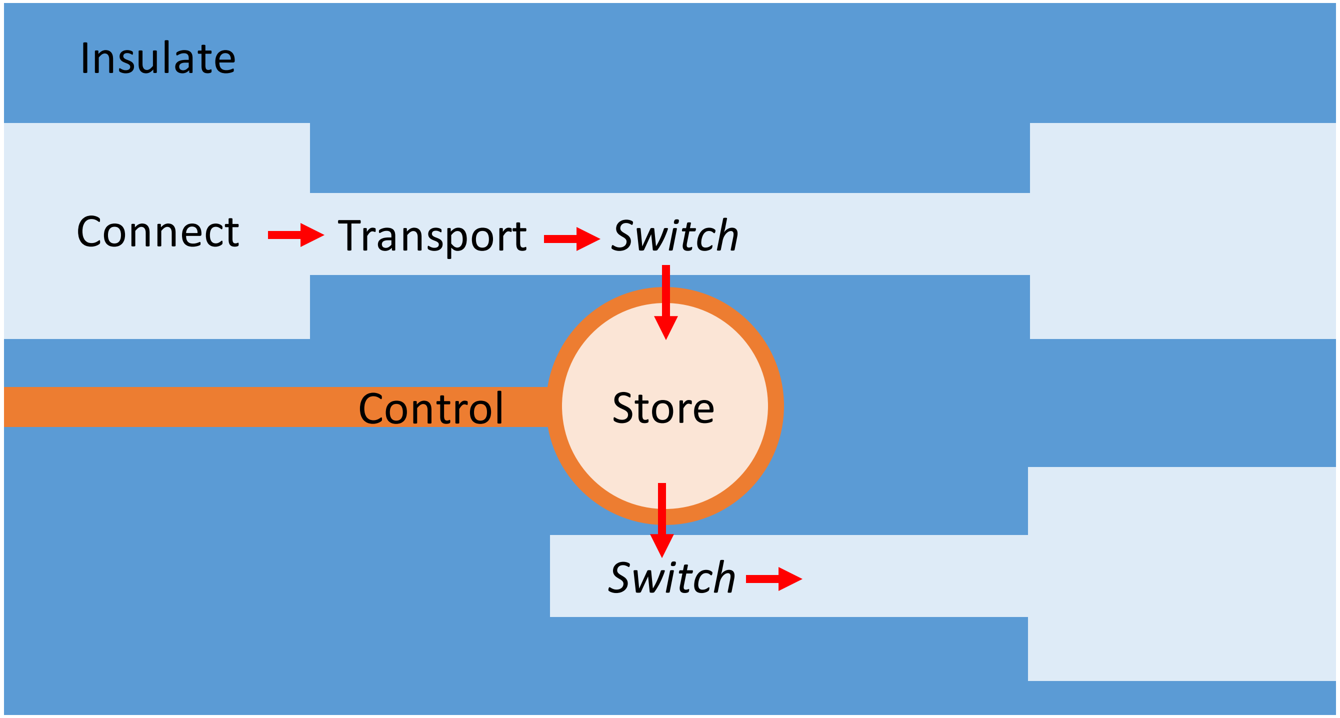


Matthew Arnold, UTS

Let’s start the same way as the previous lecture

Optical circuit analogy

Computers use miniature circuits of insulators, semiconductors & conductors



The size of photonic circuits (μm) is large compared to electronic (0.01μm)

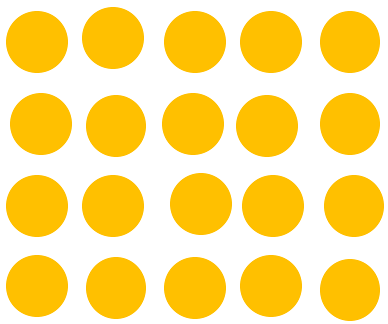
A major research goal is to combine **photons + electrons = plasmonics**

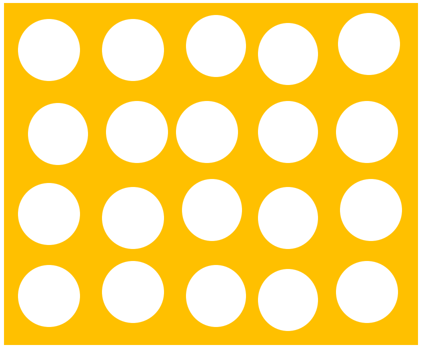
There are some obvious differences that have to be overcome

Insulator crystals

Just use electrical insulators

There *are* structures analogous to crystals – called "plasmonic **metamaterials**"
However their function is not really as insulators





Mostly transmitting at low frequencies

Conductor (diluted)
Mostly reflecting at low frequency

Both can have (very) interesting resonances
Primarily determined by individual resonators

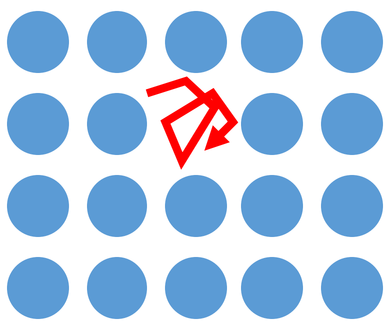
The behaviour of plasmonic metamaterials is complicated
We may return to this at a later stage

Resonator

To trap information

Different/absent inclusion

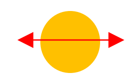
Photonic resonator



low index

Particle or disc

Plasmonic resonator



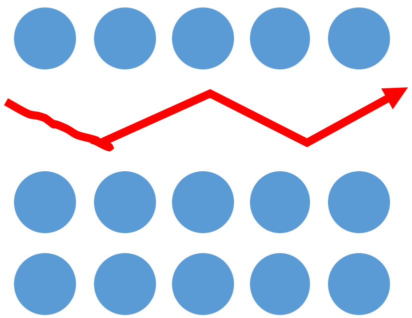
***Localized* Surface Plasmon (LSP)**

Conducting defect

To transport information

Line of missing inclusions

Photonic waveguide

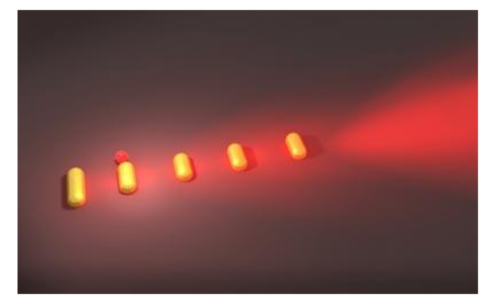


Plasmonic waveguide

film/strip-line

Surface Plasmon *Polariton* (SPP)

Optical circuits



Antenna – wireless

Conductors have highly dispersive response

Let's review some of the key points of plasma dispersion

Drude model

http://en.wikipedia.org/wiki/Drude_model

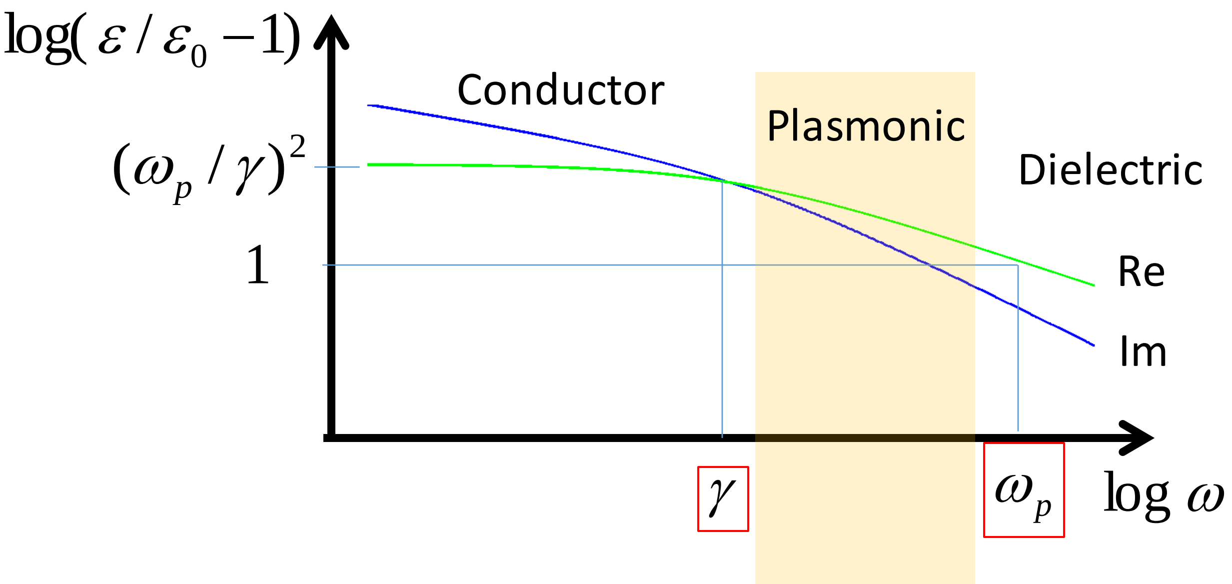
Plasma = nearly **free electrons** damped by interactions

Exists in conductors (metals AND doped semiconductors)

$$\boxed{\varepsilon/\varepsilon_0 - 1 = \frac{-\omega_p^2}{\omega^2 + i\omega\gamma}}$$

Plasma frequency ∝ charge density $\omega_p^2 = Ne^2/(m\varepsilon_0)$

Damping frequency $\gamma = \varepsilon_0 \omega_p^2 \rho_{DC}$



Negative permittivity
Some imaginary = lossy

Let's do some examples of plasma parameters for conductors

noting that Drude is generally a gross simplification,
especially near plasma frequency.

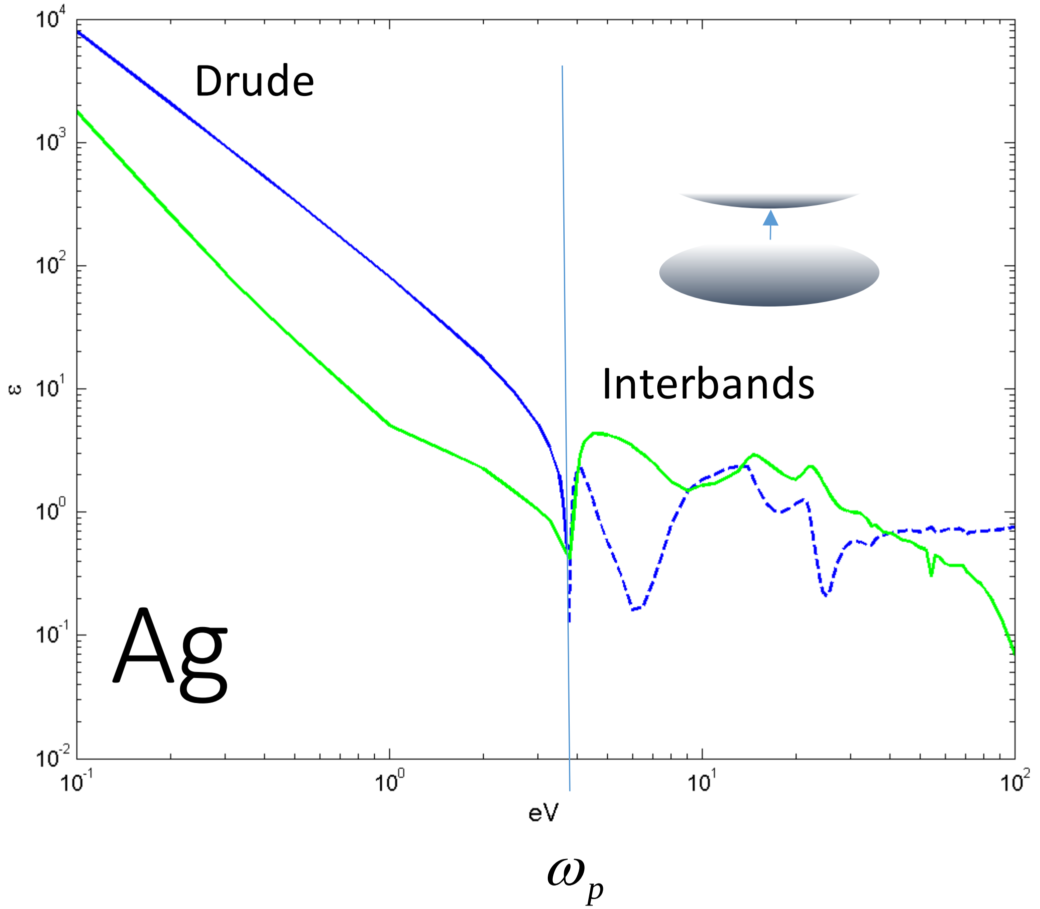
Material	**Carrier conc** cm^{-3}	**DC resist** Ω cm	**Plasma µm**	**Damping µm**
Al	$1.8\text{x}10^{23}$	3×10^{-6}	0.08	10^{1}
K	$1.4\text{x}10^{22}$	7×10^{-6}	0.3	$7\text{x}10^{1}$
low carrier-density conductors (e.g. TiN, ITO, VO_2) their properties vary depending on preparation				
Si heavy	10^{20}	10^{-3}	3	$7\text{x}10^{1}$
Si undoped	10^{12}	10^{4}	$3\text{x}10^{4}$	$7\text{x}10^{2}$

$$\lambda_p = 2\pi c / \sqrt{Ne^2/(m\varepsilon_0)}$$

$$\lambda_\gamma = \lambda_p^2 / (\varepsilon_0 \rho_{DC} 2\pi c)$$

Drude vs real material

In many real materials the "high" frequency approximation is only good when $\omega \ll \omega_p$

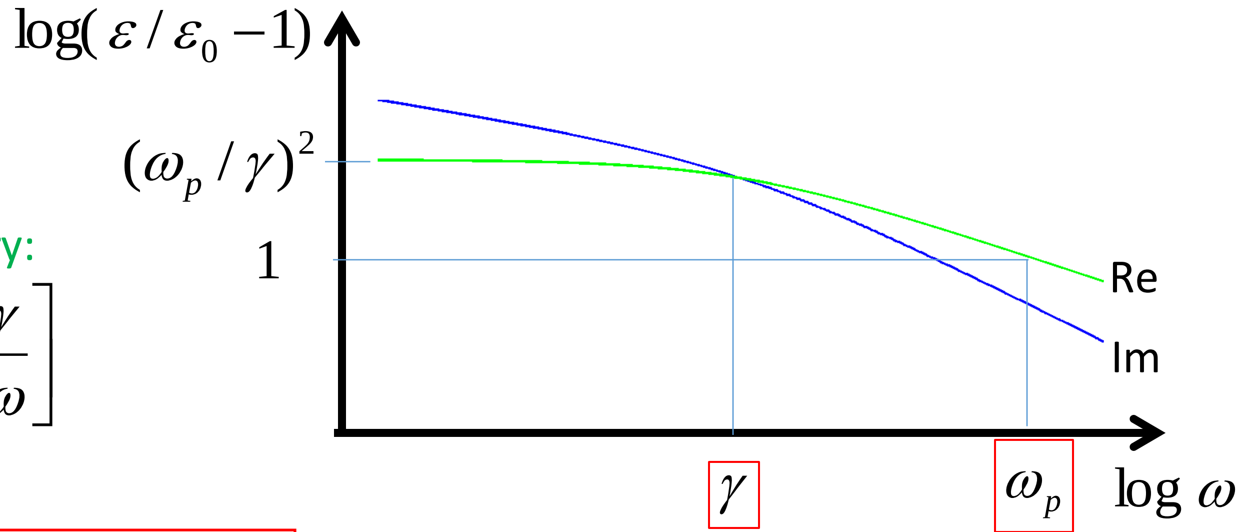


- *Estimated* plasma frequency for Ag is 9eV (140nm)
- *Observed* frequency is 4eV (340nm): due to interbands

Drude model limits

Let's look at some approximations to simplify later derivations

$$\boxed{\varepsilon / \varepsilon_0 - 1 = \frac{-\omega_p^2}{\omega^2 + i\omega\gamma}}$$



Splitting into real and imaginary:

$$\varepsilon / \varepsilon_0 - 1 = \frac{\omega_p^2}{\omega^2 + \gamma^2}\left[-1 + i\frac{\gamma}{\omega}\right]$$

Now the approximations…

Low ω << γ: $\varepsilon / \varepsilon_0 - 1 \sim \frac{\omega_p^2}{\gamma^2}\left[-1 + i\frac{\gamma}{\omega}\right]$

Imag part dominates (currents)

High ω >> γ: $\varepsilon / \varepsilon_0 - 1 \sim \frac{\omega_p^2}{\omega^2}\left[-1 + i\frac{\gamma}{\omega}\right]$

Real part dominates (displacement field)

In many real materials the "high" frequency approximation is only good when $\omega << \omega_p$

Nevertheless Drude is generally ok at moderate frequency and is easy to write down

Plasmons

Plasma = gas of free charge carriers, e.g. the carriers in a conductor

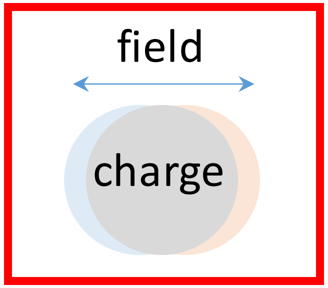
‘on’ = wavicle

A surface plasmon is a
- collective oscillation of charge carriers near a surface
- a quasiparticle hybrid between electron and photon
- an evanescent wave on a conducting surface…

There are two limiting cases of plasmons
- **LSP** = localized surface plasmon (e.g. on a small particle)
- **SPP** = surface plasmon polariton (e.g. on a metal film)

The distinction can become less obvious in-between these two cases

Localized Surface Plasmons (LSP): historical development

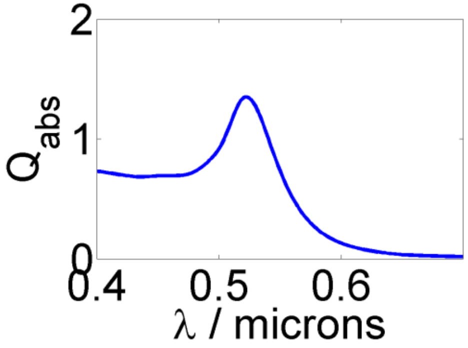


Lycurgus cup: made 4th century

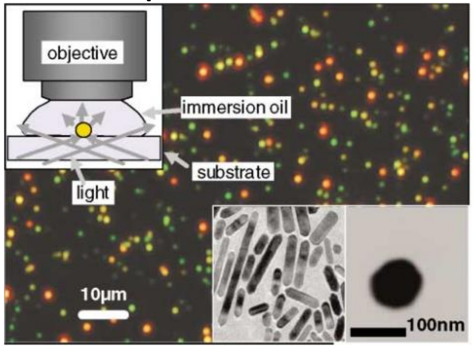
British Museum

Resonant

Optical spectroscopy:
18th & 19th centuries

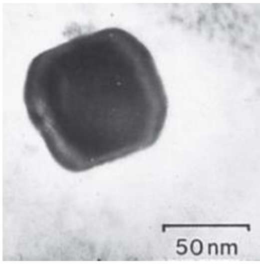


Dark-field microscopy developed 1903

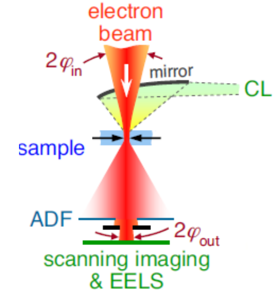


Sönnichsen et al (2002) PRL **88** 077402

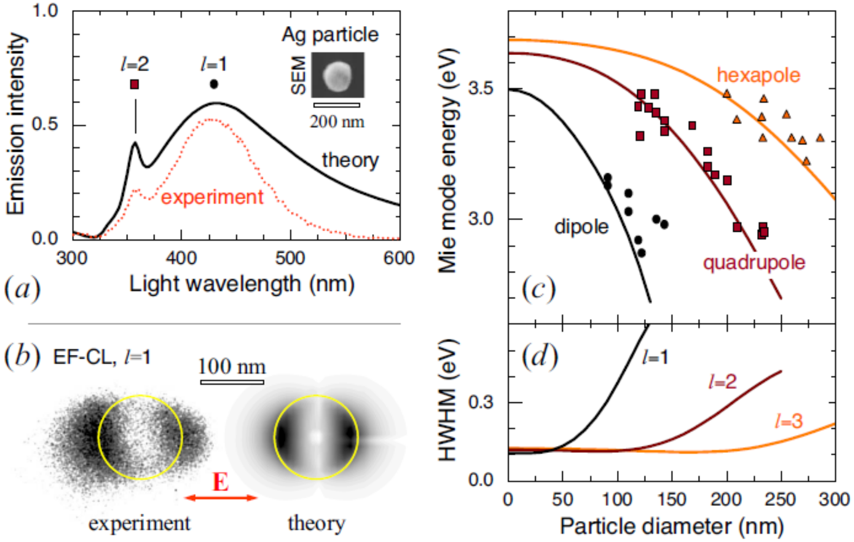
TEM (developed 1930s)



LSP nanoscale mapping

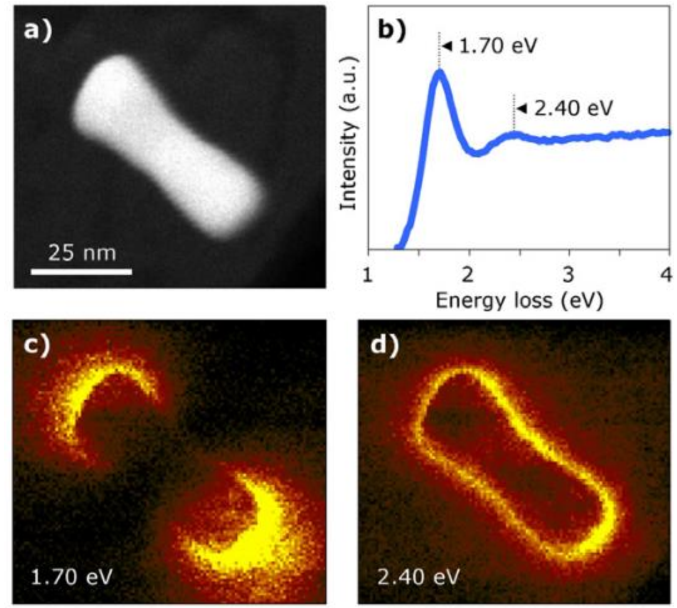


Cathodoluminescence



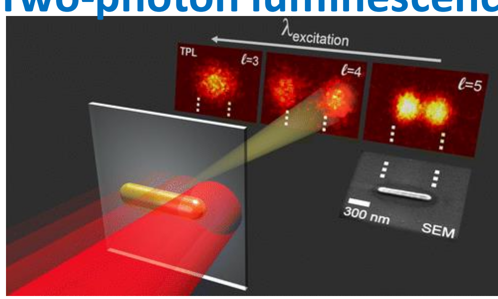
Yamamoto et al (2001), PRB **64** 205419

Electron Energy Loss



Bosman et al (2007) Nanotech **18** 165505

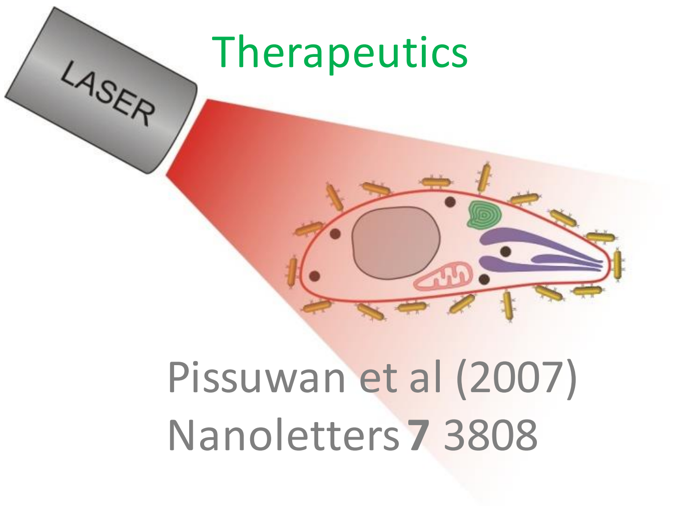
Two-photon luminescence



ACS Photonics, 2015, 2, 410

Localized Surface Plasmons: applications

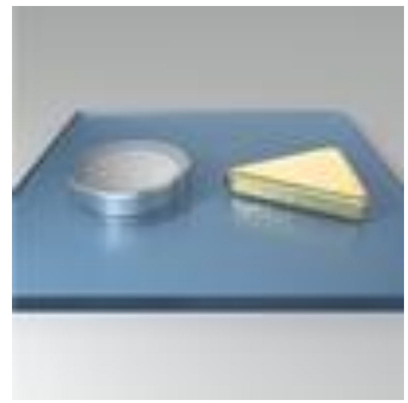
Therapeutics



Pissuwan et al (2007)
Nanoletters **7** 3808

1. Antibody coated particle sticks to target
2. Laser passes through body
3. Particles absorb → heat → localized kill

Sensing



dx.doi.org/10.1364/OME.2.000111

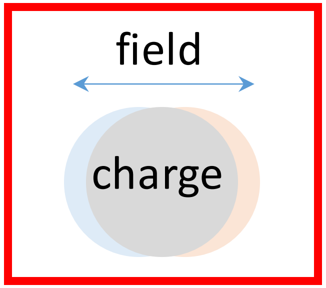
1. Analyte changes local refractive index
2. LSP peak shifts
3. Detected

Example above:
Pd disc changes n when absorbs H
Au triangle resonance shifts

Understanding LSP – charge model

(because we want to know to control frequency etc)

- Drude (free e-) model explains conductor response
- Excitation field forces opposite charge to boundaries
- Separated charge provides restoring force
- Looks like bound charge (surface) → Lorentz oscillator
- Boundary shape → distribution → force → energy



Quasistatic model

(Lorentz oscillator)

- <u>Assumes particle is so small that</u>
  - "sloshing" time is negligible
  - field penetrates fully through particle
  - response mainly due to internal displacement field (not currents)
- Geometry and material properties "separable"
- Maximum response at resonance ($1/\chi \approx -S$)
- **Limiting factor**: χ is complex

$$\alpha = V \sum_m \frac{A_m}{S_m + 1/\chi} \qquad 1/\chi \equiv \frac{1}{\varepsilon/\varepsilon_b - 1}$$

A: Strength (0 to 1)

ε_b = background

Size

Shape

Material

S: Position (0 to 1)

Sphere dipole=1/3

Rod dipoles: transverse→1/2, long→0

Disc dipoles: plane→0, normal→1

LSP: effect of metal

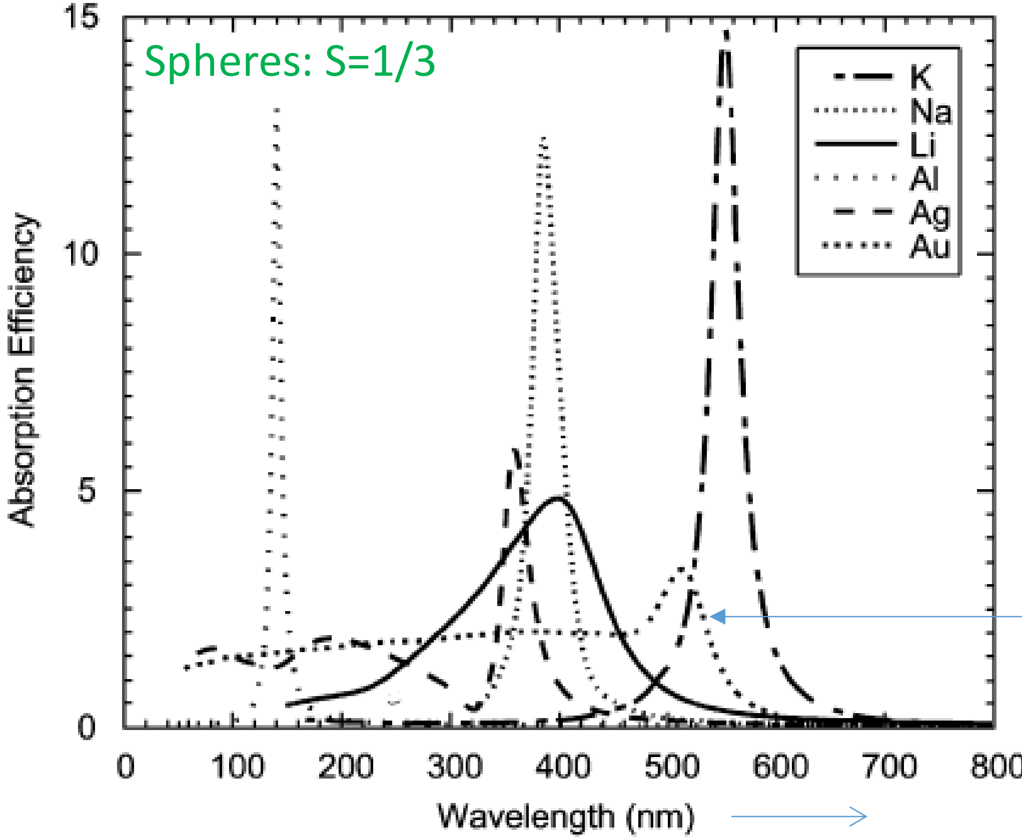
Position explained by factors affecting valence density

Height mostly optical resistivity

Al: $3s^2$ $3p^1$ Lots of free e-

Na: $3s^1$

K: $4s^1$ Big cloud



Au: $4f^{14}$ $5d^{10}$ $6s^1$

Complicated orbits

Electron scattering likely

Where is the dipole resonance of small metal spheres?

$$\alpha_{sphere} \propto \frac{1}{1/3+1/(\varepsilon/\varepsilon_b - 1)}$$

Note: unlike photonic resonance
LSP position is nearly independent of size

α max when denom near zero: $\boxed{\varepsilon_{sphere}/\varepsilon_b \approx -2}$

For Drude (<u>inaccurate</u> for many materials near sphere resonance): $\varepsilon/\varepsilon_0 \sim 1-\frac{\omega_p^2}{\omega^2}$

$$\boxed{\omega_{sphere} \sim \omega_p/\sqrt{1+2\varepsilon_b/\varepsilon_0}}$$

A high index background (e.g. water or glass) redshifts the resonance from free-space

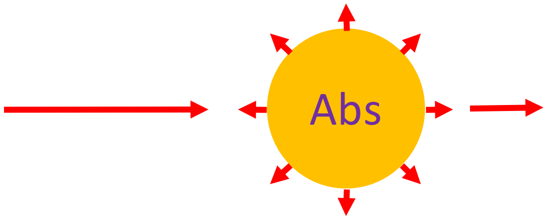
In free-space: $\omega_{sphere,0} \sim \omega_p/\sqrt{3}$

What's the approx. resonant wavelength of a small potassium sphere in vacuum?
How about in glass?

Far-field LSP response

How is energy divided?

Input



Transmitted Direct beam

Measurements

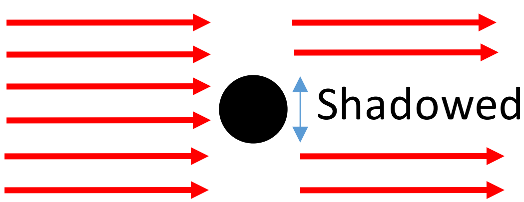
Scatter
Integrated outgoing

Energy balance: Transmission + $\boxed{\text{Scatter + Absorption}}$ = Input

Extinction

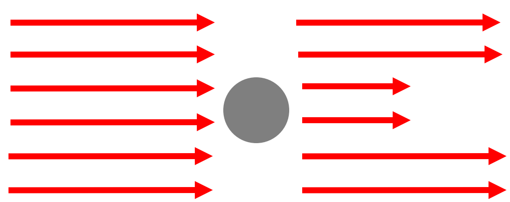
Generally expressed as cross-sections, i.e. the equivalent "area"

Perfect **large** absorber



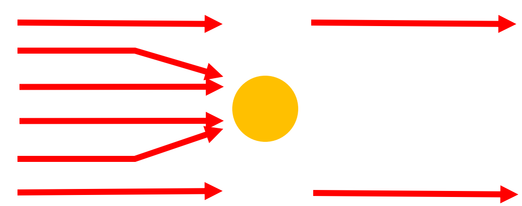
$C = \pi r^2$

Weak absorber



$C < \pi r^2$

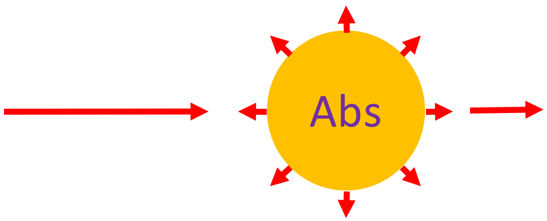
LSP can absorb **more** than area!



$C > \pi r^2$

Far-field LSP response

How is energy divided?

Input 

Transmitted Direct beam

Measurements

Scatter

Integrated outgoing

Energy balance: Transmission + $\boxed{\text{Scatter + Absorption}}$ = Input

Extinction

Cross-sections from (dipole) polarizability

Small particle limit:

$$C_{abs} \approx k\,\mathrm{Im}\{\alpha\}$$

$$k = 2\pi / \lambda$$

$$C_{sca} \approx \frac{\pi}{6} k^4 |\alpha|^2$$

(Rayleigh scattering)

Resonant cross-sections

$$\alpha_{ellipsoid} = V \frac{1}{S + 1/\chi}$$

At resonance, S cancels negative real part of 1/χ, leaving just the imaginary part

Drude
(inaccurate for near sphere resonance):

$$\varepsilon_{\Re} - 1 \sim -\frac{\omega_p^2}{\omega^2} \qquad \varepsilon_I \sim \frac{\gamma\omega_p^2}{\omega^3}$$

$$\boxed{\alpha_{resonance} \sim iV(\tilde{\varepsilon}_{\Re} - 1)^2 / \tilde{\varepsilon}_{\Im}} \rightarrow \boxed{\alpha_{resonance} \sim iV \frac{\omega_p^2}{\gamma\omega}}$$

$\varepsilon_{\Im} << |\varepsilon_{\Re}|$ assuming low loss

$$C_{abs} \approx kV(\tilde{\varepsilon}_{\Re} - 1)^2 / \tilde{\varepsilon}_{\Im}$$

$$C_{abs} \approx V \frac{\omega_p^2}{\gamma c}$$

Independent of ω

$$C_{sca} \approx \frac{\pi}{6} k^4 \left[V(\tilde{\varepsilon}_{\Re} - 1)^2 / \tilde{\varepsilon}_{\Im} \right]^2$$

$$C_{sca} \approx \frac{\pi}{6} \left[\frac{V\omega\omega_p^2}{\gamma c^2} \right]^2$$

Increases faster with V

Resonant cross-sections: sphere

$$\boxed{\alpha_{sphere} \sim iV9/\tilde{\varepsilon}_{\Im}}$$

$$\boxed{\alpha_{sphere} \sim iV\frac{\omega_p\sqrt{3}}{\gamma}}$$

$$C_{abs} \approx kV9/\tilde{\varepsilon}_{\Im}$$

$$C_{abs} \approx V\frac{\omega_p^2}{\gamma c}$$

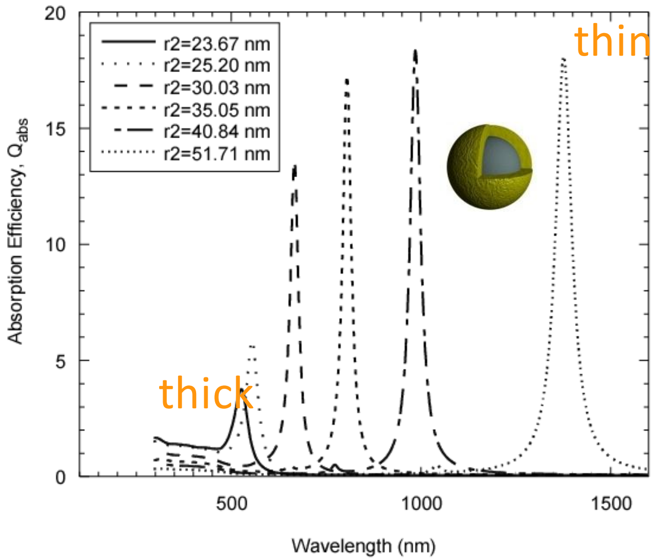
$$C_{sca} \approx \frac{\pi}{6}k^4\left[V9/\tilde{\varepsilon}_{\Im}\right]^2$$

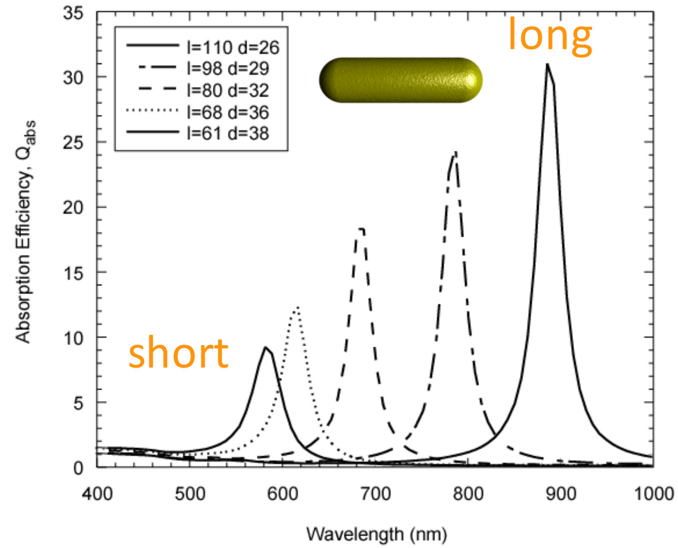
$$C_{sca} \approx \frac{\pi}{6}\left[\frac{V\omega_p^3}{\gamma c^2\sqrt{3}}\right]^2$$

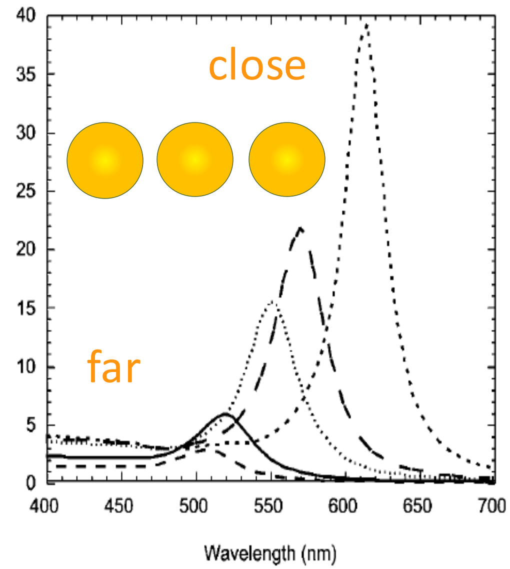
What is C_{abs} for a K sphere of 10nm radius?
Compare to the geometric cross-section.

Effect of shape on LSP

*Sensitive to **shape***







Gold Bulletin **41** 5

S=1/3 → S→0

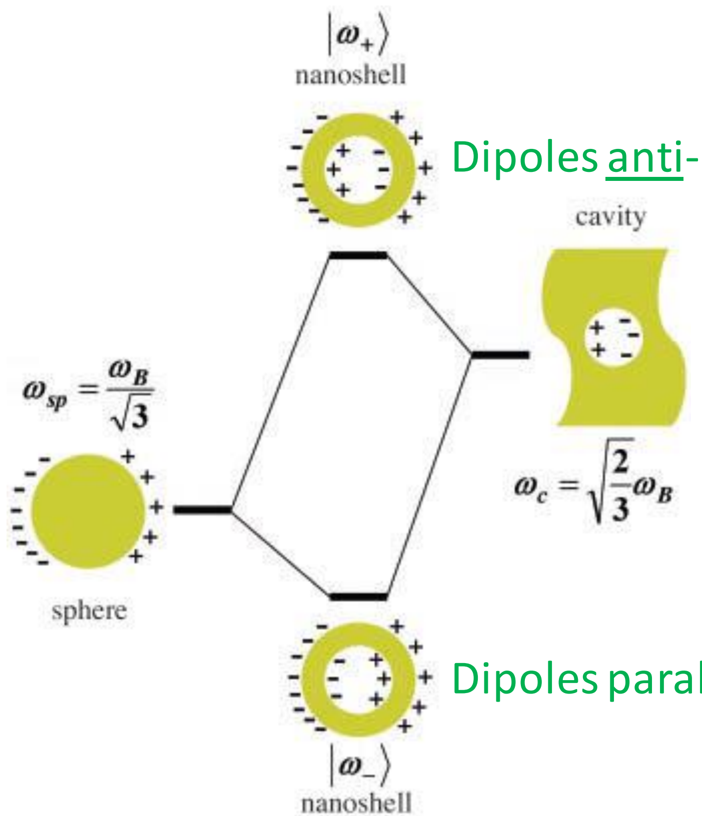
JPhysChemC **113**, 2784

Notice gets better at long λ
Primarily due to Au short λ interband absorb

Now: let's try to understand position in terms of charge distribution

Hybridization of core-shells

Dipole modes of sphere & cavity overlap → energy splitting

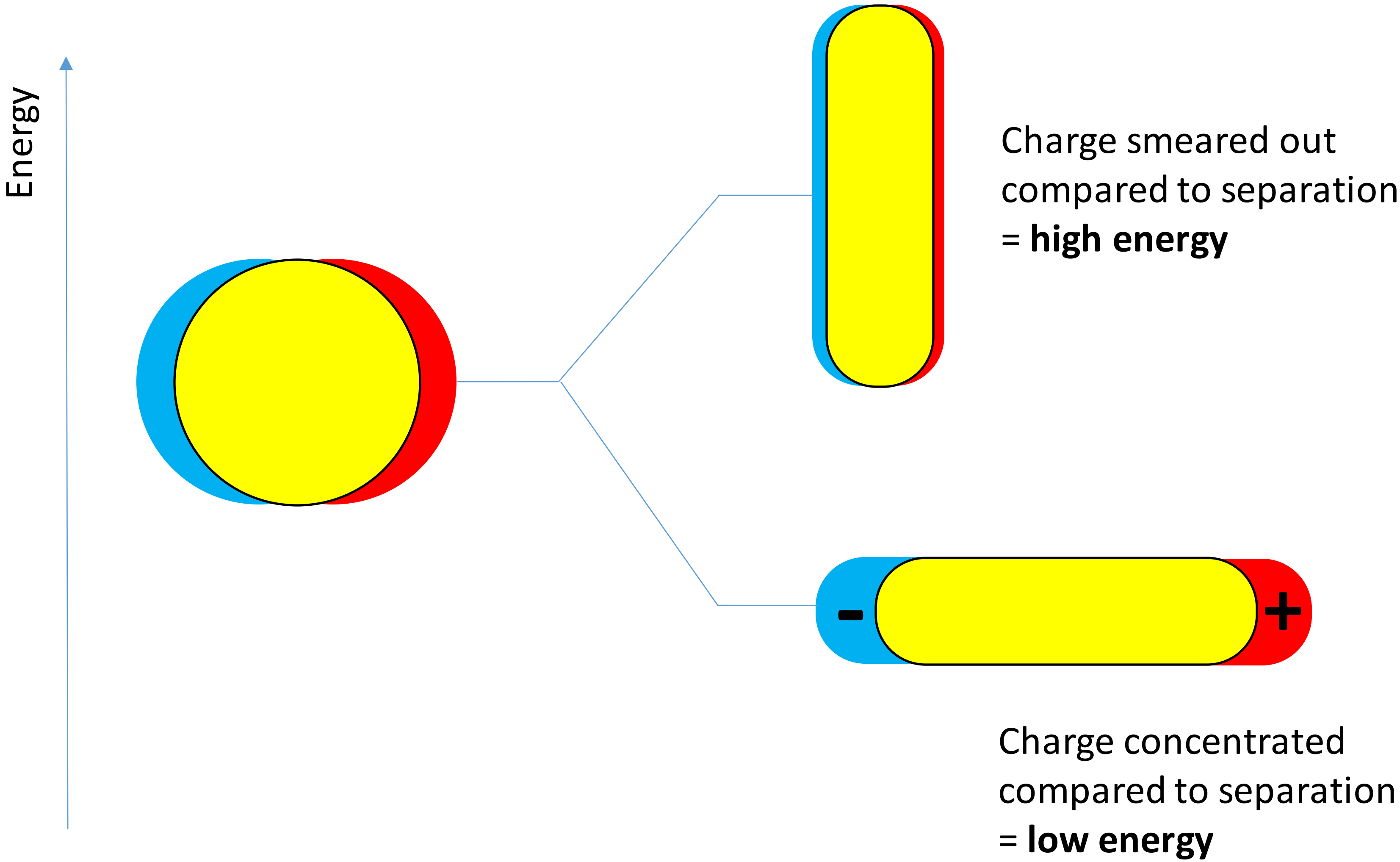


Dipoles anti-parallel: weakly radiative

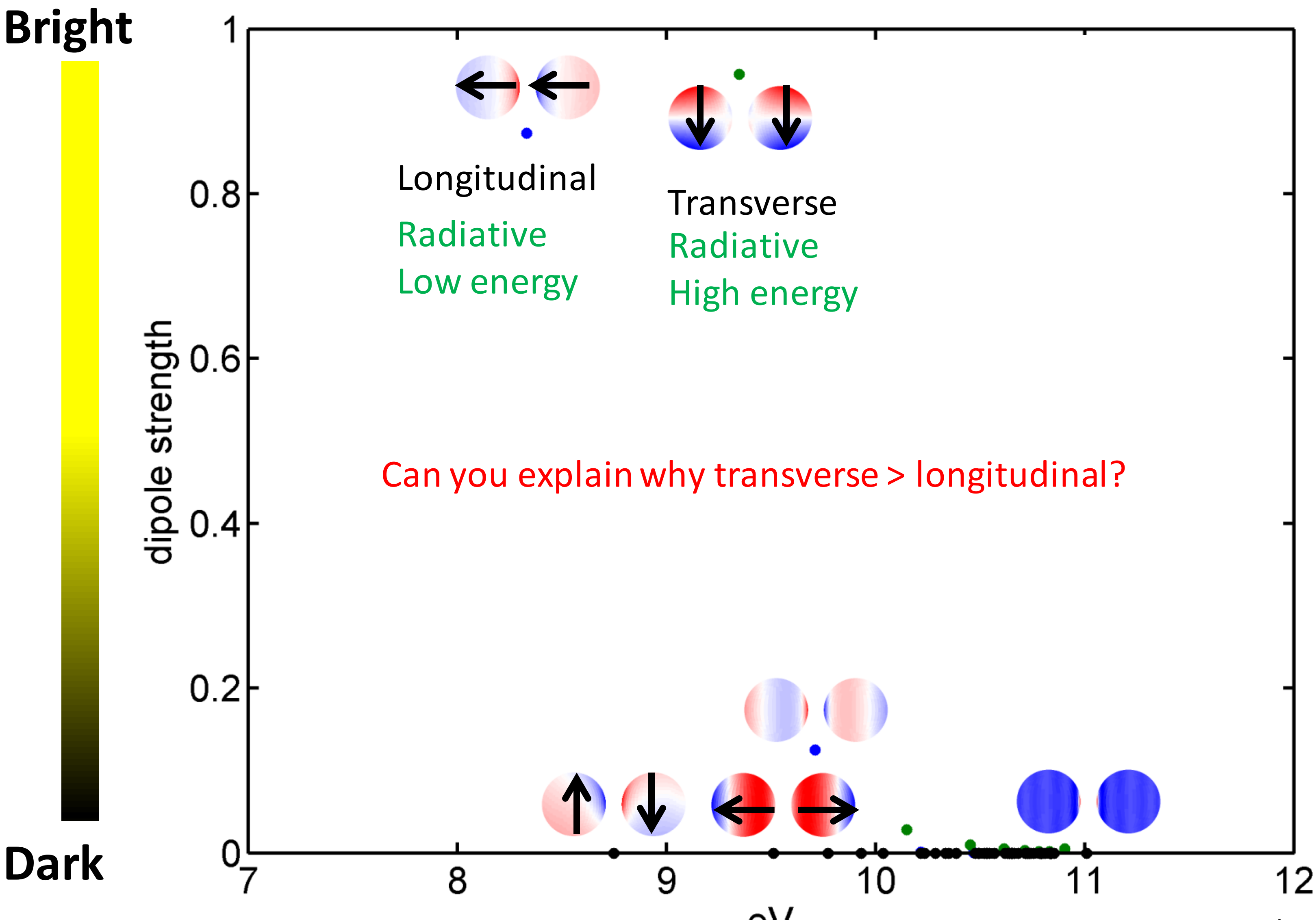
Dipoles parallel: radiative

Science 302, 419

Shift of rod resonances



Hybridization of sphere dimer



(ω_p=15.8eV)

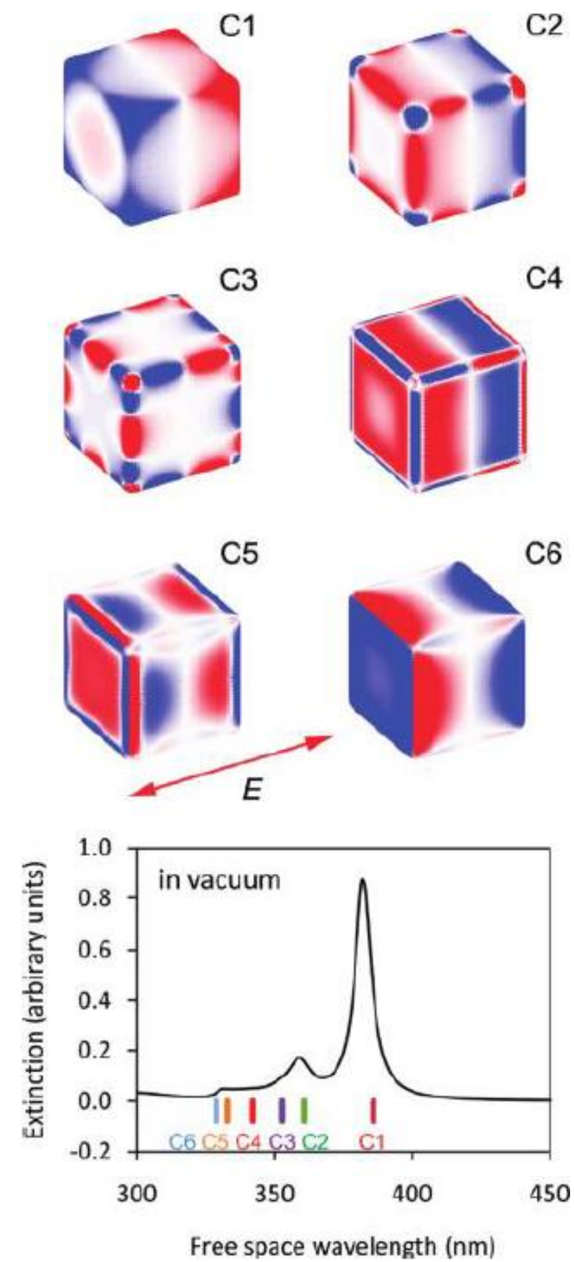
Higher order modes

So far we have only considered dipole modes

Higher order modes have more complicated distribution
- Higher energy
- Less radiative

Even ellipsoidal shapes have higher order modes
But these are not radiative until the particle becomes large

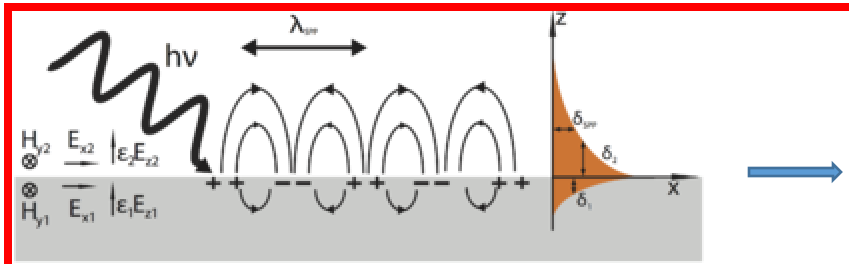
Other shapes can have very interesting modes



Langmuir 28 9103

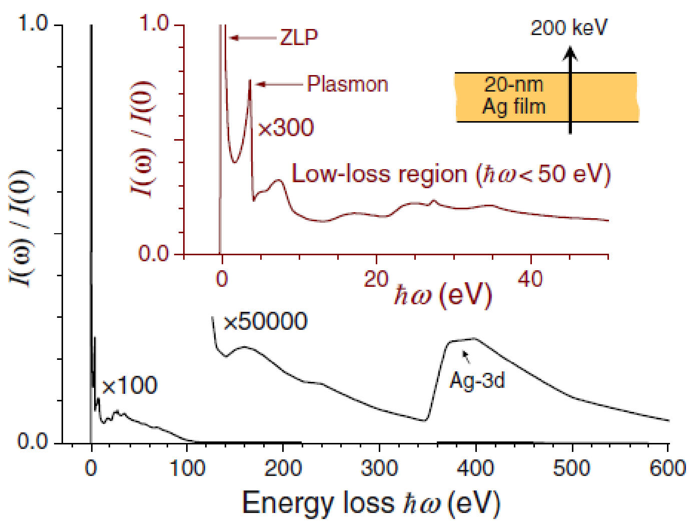
Surface plasmon polariton (SPP)

Predicted and observed around 1960s



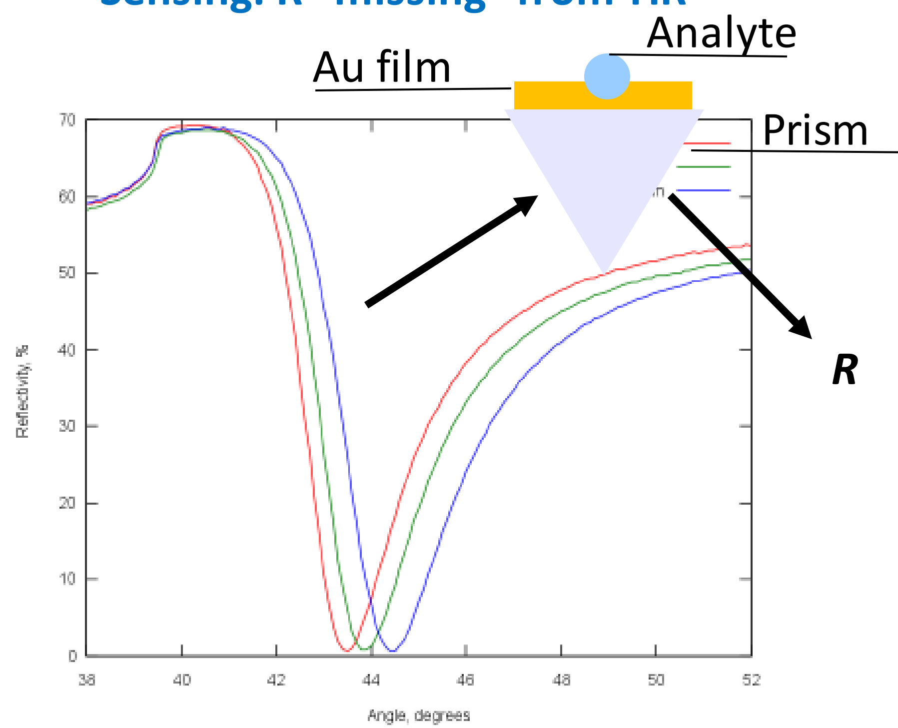
Travelling wave

Electron transmission loss

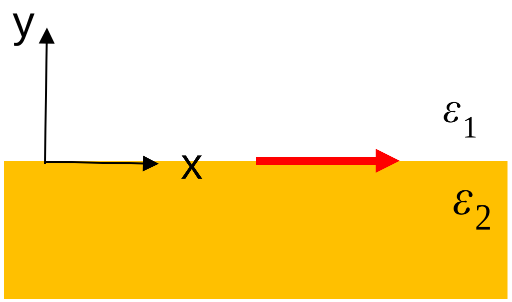


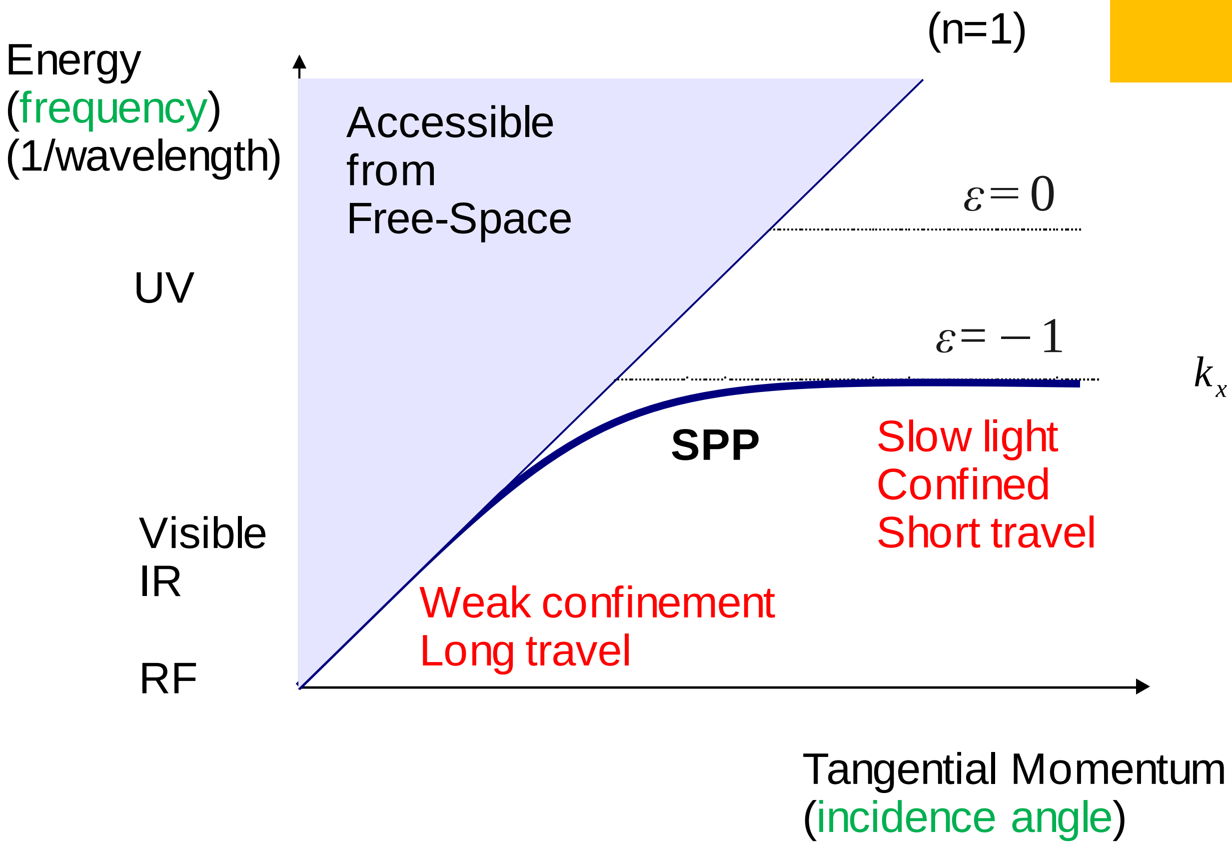
RevModPhys 82 209

Sensing: R "missing" from TIR



Dispersion - SPP



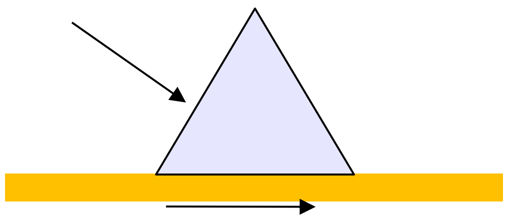


$$k_x = \frac{\omega}{c}\sqrt{\frac{\tilde{\varepsilon}_1\tilde{\varepsilon}_2}{\tilde{\varepsilon}_2+\tilde{\varepsilon}_1}}$$

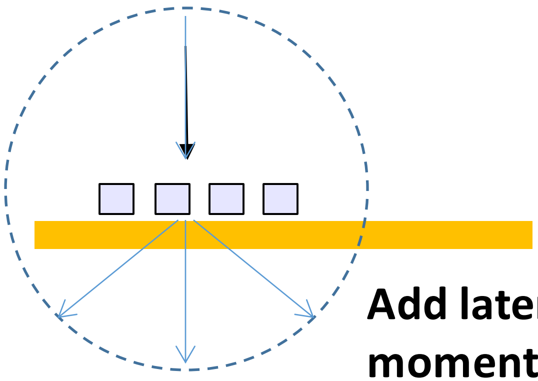
SPP doesn't intersect light-line = no coupling

Coupling

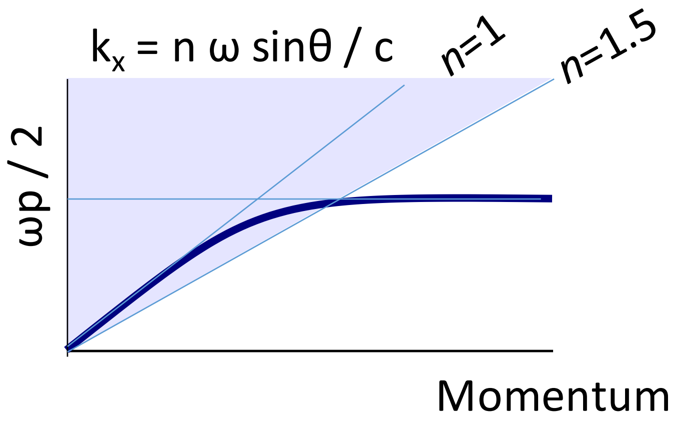
Need to address momentum mismatch



Move the light-line

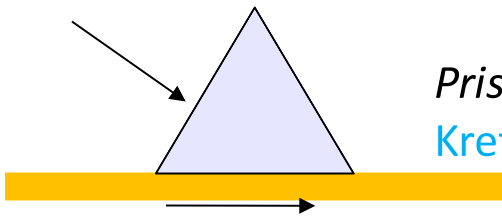


Add lateral momentum



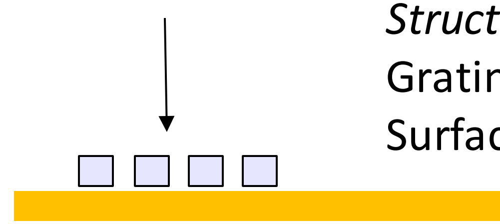
If n=1.5, what's the *minimum* internal angle to couple to SPP?

Coupling



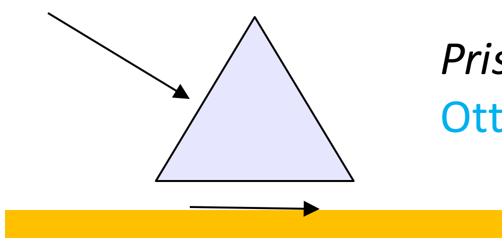
Prism
Kretschmann

Move the light-line



Structure
Grating
Surface roughness

Add lateral momentum



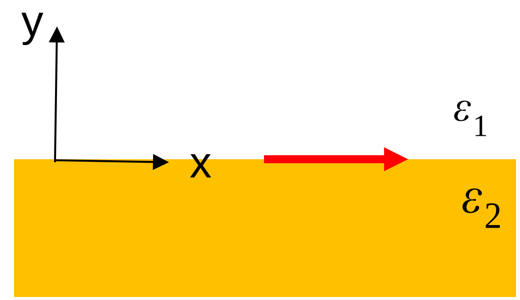
Prism
Otto

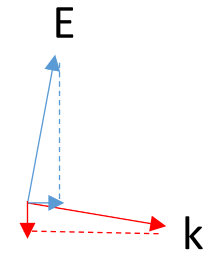
Move the light-line

Narrow beam
NSOM
Electron beam (input only)

High momentum probe

(Short) derivation of mode





Two media
Planes waves “p” polarization
“No” incident wave = trapped mode very large

Projection of orthogonal k & E
$$E_{1x} = \frac{k_{1y}}{k_1} E_1 = \sqrt{1-\left(\frac{k_x}{k_1}\right)^2} E_1$$

Transverse EM
$$E_{1x} = \sqrt{1-\left(\frac{k_x}{\sqrt{\varepsilon_1 \mu_1}\,\omega}\right)^2} \sqrt{\frac{\mu_1}{\varepsilon_1}} H_1$$

Maxwell
Boundary cond.
$$E_{1x} = E_{2x} \qquad H_1 = H_2$$

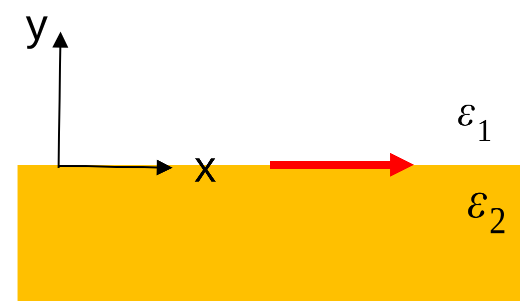
Substitute
& non-mag.
$$\sqrt{1-\left(\frac{k_x}{\sqrt{\varepsilon_1 \mu_0}\,\omega}\right)^2} \sqrt{\frac{\mu_0}{\varepsilon_1}} = \sqrt{1-\left(\frac{k_x}{\sqrt{\varepsilon_2 \mu_0}\,\omega}\right)^2} \sqrt{\frac{\mu_0}{\varepsilon_2}}$$

Square,
rearrange
$$(\mu_0 \omega^2 \varepsilon_1 - k_x^2)\,\varepsilon_2^2 = (\mu_0 \omega^2 \varepsilon_2 - k_x^2)\,\varepsilon_1^2$$

Solve for k
$$k_x = \omega \sqrt{\mu_0 \frac{\varepsilon_1 \varepsilon_2^2 - \varepsilon_2 \varepsilon_1^2}{\varepsilon_2^2 - \varepsilon_1^2}} = \omega \sqrt{\mu_0 \frac{\varepsilon_1 \varepsilon_2}{\varepsilon_2 + \varepsilon_1}} = \frac{\omega}{c} \sqrt{\frac{\tilde{\varepsilon}_1 \tilde{\varepsilon}_2}{\tilde{\varepsilon}_2 + \tilde{\varepsilon}_1}}$$

Relative
permittivities

Interpretation



$$k_x = \frac{\omega}{c}\sqrt{\frac{\tilde{\varepsilon}_1\tilde{\varepsilon}_2}{\tilde{\varepsilon}_2+\tilde{\varepsilon}_1}}$$

Tangential k: want real (propagating)

$$k_{1y} = \sqrt{k_1^2 - k_x^2} = \frac{\omega}{c}\sqrt{\tilde{\varepsilon}_1 - \frac{\tilde{\varepsilon}_1\tilde{\varepsilon}_2}{\tilde{\varepsilon}_1+\tilde{\varepsilon}_2}} = \frac{\omega}{c}\frac{\tilde{\varepsilon}_1}{\sqrt{\tilde{\varepsilon}_1+\tilde{\varepsilon}_2}}$$

Normal: want imag (trap)

* Only opposite sign permittivities give evanescent waves
* This is provided by metal:dielectric combination
* Normal part gives us confinement (large imaginary k means tight confinement)
* Tangential part gives propagation length (must allow complex permittivities)...

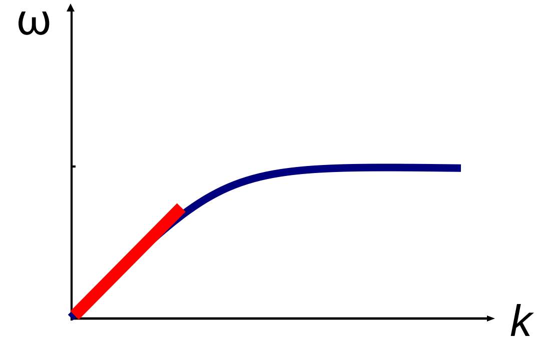
$\varepsilon_2 = \varepsilon_2' + i\varepsilon_2''$

$$k_x = \frac{\omega}{c}\sqrt{\frac{\tilde{\varepsilon}_1\tilde{\varepsilon}_2}{\tilde{\varepsilon}_1+\tilde{\varepsilon}_2}} = \frac{\omega}{c}\sqrt{\frac{\tilde{\varepsilon}_1\tilde{\varepsilon}_2'}{\tilde{\varepsilon}_1+\tilde{\varepsilon}_2'}}\left[1 + i\frac{\tilde{\varepsilon}_1\tilde{\varepsilon}_2''}{2(\tilde{\varepsilon}_1+\tilde{\varepsilon}_2')\tilde{\varepsilon}_2'}\right]$$

Assuming:
Small imaginary permittivity

Small imaginary wave-vector
Long propagation length
Light like

Drude: light-like



$$k_x = \frac{\omega}{c}\sqrt{\frac{\tilde{\varepsilon}_1\tilde{\varepsilon}_2}{\tilde{\varepsilon}_1+\tilde{\varepsilon}_2}} = \frac{\omega}{c}\sqrt{\frac{\tilde{\varepsilon}_1\tilde{\varepsilon}_2'}{\tilde{\varepsilon}_1+\tilde{\varepsilon}_2'}}\left[1+i\frac{\tilde{\varepsilon}_1\tilde{\varepsilon}_2''}{2(\tilde{\varepsilon}_1+\tilde{\varepsilon}_2')\tilde{\varepsilon}_2'}\right]$$

$$\tilde{\varepsilon}_2 \approx -\frac{\omega_p^2}{\omega^2}+i\frac{\omega_p^2\gamma}{\omega^3} \qquad \gamma << \omega << \omega_p \qquad \tilde{\varepsilon}_1 \to 1$$

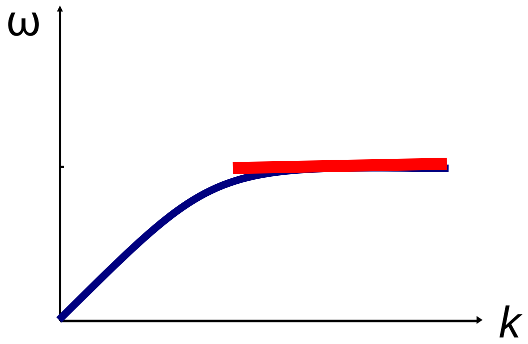
Light-like

$$k_x \sim \frac{\omega}{c}\left[1+i\frac{\gamma\omega}{2\omega_p^2}\right]$$

The imaginary part is not very accurate unless

$$\omega << \omega_p$$

Drude: slow light



$$\omega \sim \omega_p / \sqrt{2}$$

Slow light

$$\tilde{\varepsilon}_2 \approx -1 + i\,\frac{2\sqrt{2}\gamma}{\omega_p}$$

We can’t use the previous expression for k (here is too lossy)…
do series expansion, then *very* approximately

$$k_x \sim \frac{\omega_p}{c2^{7/4}\sqrt{\gamma/\omega_p}}\left[1+i\right]$$

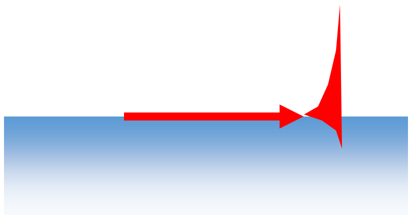
$$k_x \sim \frac{\omega}{c}\left[1 + i\,\frac{\gamma\omega}{2\omega_p^2}\right] \qquad\qquad k_x \sim \frac{\omega_p}{c\,2^{7/4}\sqrt{\gamma/\omega_p}}\left[1+i\right]$$

Estimate the propagation length 1/Im(k) for Al @ 1.55µm and at the “slow light” limit

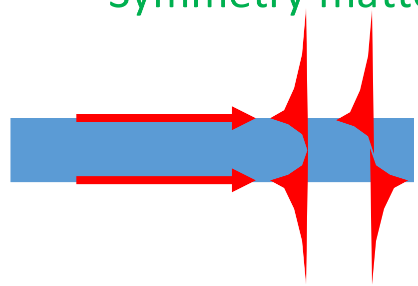
Thin films stacks

Plasmons exist at each surface: they hybridize

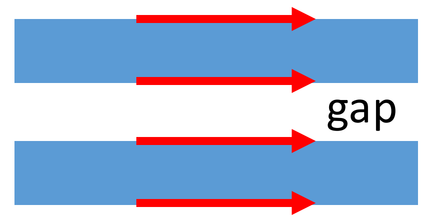
Symmetry matters



Single frequency



Film: two frequencies



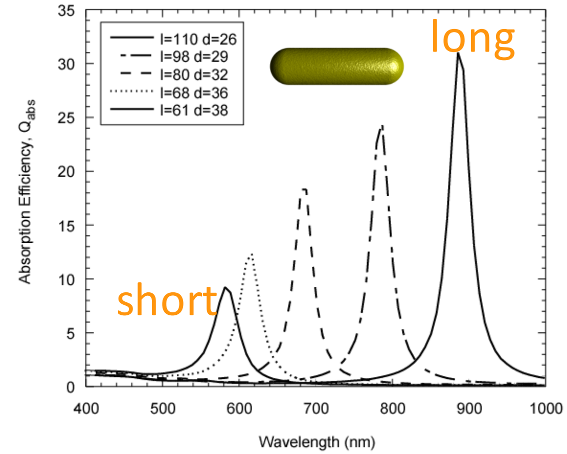
Four frequencies

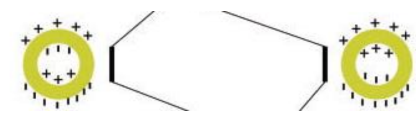
The concentration of field in the metal affects the loss

If we can trap the plasmon in a **gap** it should have lower loss

Summary

- Plasmons = surface resonance of conductor
- Drude model useful for basic material response
- LSP: small particle traps
  - Shape most useful for tuning
  - Electron forces key to understanding
  - Modelled by quasistatic resonance
- SPP: thin film waveguides
  - Needs special coupling (prism/gratings)
  - Highly dispersive: light-like → slow light
  - Thickness determines properties





$$\alpha = V \sum_m \frac{A_m}{S_m + 1/\chi}$$